



Types of Data to Be Mined

INTRODUCTION TO DATA MINING

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What is Data Mining?

- ▶ - Extracts meaningful patterns and trends from raw data
- ▶ - Supports strategic business decisions and predictions
- ▶ - Involves statistics, machine learning, and database systems
- ▶ - Common tasks: classification, clustering, regression, and association rule mining
- ▶ - Real-life Examples: Google search predictions, Amazon product suggestions, bank fraud alerts

Why Types of Data Matter?

- ▶ - Data can be structured, semi-structured, or unstructured
- ▶ - Each data type requires a unique preprocessing and mining approach
- ▶ - Impacts the choice of storage system, mining algorithm, and interpretation method
- ▶ - Example: Text data requires NLP, while tabular data uses SQL or statistical methods

Structured Data

- ▶ - Clearly defined format, typically in rows and columns (relational tables)
- ▶ - Easily stored and queried using SQL-based systems (e.g., MySQL, PostgreSQL)
- ▶ - Highly reliable and easy to analyze due to consistency
- ▶ - Mining methods include decision trees, regression models, and clustering algorithms
- ▶ - Example: Analyzing monthly sales data to identify best-selling products

Semi-Structured Data

- ▶ - Contains tags or markers to separate semantic elements but lacks a rigid schema
- ▶ - Can be partially queried and processed with flexible tools (e.g., XQuery, JSONPath)
- ▶ - Common in data exchanged between systems (e.g., APIs, config files)
- ▶ - Requires parsing before applying traditional mining algorithms
- ▶ - Example: Parsing and analyzing review data from e-commerce websites

Unstructured Data

- ▶ - No predefined format; cannot be stored in traditional databases easily
- ▶ - Requires transformation before analysis (e.g., converting text to tokens or images to arrays)
- ▶ - Examples: Email bodies, tweets, scanned documents, customer calls
- ▶ - Advanced techniques like NLP and computer vision (CNNs, Transformers) are used
- ▶ - Example: Classifying customer feedback as positive, negative, or neutral

Time-Series Data

- ▶ - Data points collected over consistent time intervals
- ▶ - Exhibits temporal patterns such as trends, cycles, or seasonality
- ▶ - Requires specialized analysis tools that account for time-dependence (e.g., time-lag)
- ▶ - Widely used in finance, meteorology, and healthcare
- ▶ - Example: Predicting stock price movement based on historical trends

Transactional Data

- ▶ - Generated through business or system transactions (e.g., online purchases)
- ▶ - Typically involves user ID, product ID, quantity, timestamp
- ▶ - Useful for pattern discovery, such as frequent itemsets and customer behavior analysis
- ▶ - Algorithms: Apriori, ECLAT, FP-Growth for market basket analysis
- ▶ - Example: Finding product combinations frequently bought together

Spatial Data

- ▶ - Associated with geographic locations or spatial objects
- ▶ - Includes coordinates, maps, satellite imagery, and spatial networks
- ▶ - Applications: route planning, agriculture monitoring, urban planning
- ▶ - Spatial data mining involves topological and distance-based analysis
- ▶ - Example: Identifying regions with high disease incidence using GPS-tagged data

Multimedia Data

- ▶ - Includes text, audio, images, and videos that require different formats and processing methods
- ▶ - Techniques involve signal processing, feature extraction, deep learning (CNNs, RNNs)
- ▶ - Applications: facial recognition, content recommendation, image classification
- ▶ - Requires huge computational power and storage systems
- ▶ - Example: Classifying images from a traffic camera to detect violations

Web Data



- ▶ - Data derived from the web, including HTML content, logs, social media, and search queries
- ▶ - Techniques: Web scraping, web usage mining, web structure mining
- ▶ - Helps analyze user preferences, behavior trends, and page rankings
- ▶ - Used heavily in digital marketing and recommendation engines
- ▶ - Example: Analyzing clickstream data to optimize website design

Sensor/IoT Data

- ▶ - Comes from sensors embedded in smart devices (e.g., fitness trackers, smart meters)
- ▶ - Typically streaming and time-dependent; may include structured and unstructured elements
- ▶ - Requires edge computing, stream processing, and anomaly detection techniques
- ▶ - Example: Predictive maintenance in manufacturing by analyzing vibration and temperature data

Comparison of Data Types

- ▶ - Structured: Tabular, easy to query (e.g., sales database)
- ▶ - Semi-Structured: Tagged data with irregular schema (e.g., JSON/XML)
- ▶ - Unstructured: No predefined format (e.g., video, audio)
- ▶ - Time-Series: Chronologically ordered data (e.g., stock prices)
- ▶ - Transactional: Sequence of purchases or events (e.g., shopping cart logs)

Challenges in Data Mining

- ▶ - Volume: Gigabytes to petabytes of data generated daily
- ▶ - Variety: Different data formats, sources, and quality
- ▶ - Velocity: High-speed data streams require real-time processing
- ▶ - Veracity: Data can be noisy, incomplete, or biased
- ▶ - Integration: Combining data from multiple heterogeneous sources is complex

Interactive Activity

- ▶ - Match each real-world example with its data type:
- ▶ - ECG Report → Time-Series
- ▶ - Customer Tweet → Unstructured
- ▶ - SQL Table with Orders → Structured
- ▶ - Traffic Video Clip → Multimedia
- ▶ - Smartwatch Heart Rate Data → Sensor/Time-Series
- ▶ - Discussion: How would you preprocess and analyze each of these?

Summary

- ▶ - Data comes in multiple formats, each requiring unique mining strategies
- ▶ - Understanding data types is the first step to successful data mining
- ▶ - Enables better model selection, feature engineering, and interpretation
- ▶ - Real-world applications span finance, healthcare, marketing, logistics, and more